

Linear Regression Project

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BUS 110

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Introduction

Academic success is a very important aspect of development in young adults. While there are many different indicators of academic success, GPA (grade point average) is one of the most common indicators used. This project aims to seek the extent to which multiple factors, including age, study time, absences, participation in tutoring, and parental support, impact GPA. More specifically, multiple linear regression analysis on the aforementioned factors will be used to predict what factors are the most impactful on student GPA.

Data Section

The data for this project was found on kaggle.com, a subsidiary to Google. It is an internet platform used to explore, analyze, and share quality data. The official dataset is titled “Students Performance Dataset: *Academic Success Factors in High School Students*”. This data set includes 16 variables total taken from a group of 2392 students, however, only 6 will be used in this project. These are the aforementioned variables in the introduction section; age, study time, absences, participation in tutoring, and parental support, and GPA. Age is used to explore the possibility of maturity impacting student performance, and has a range from 15 years old to 18 years old. The next variable is study time, and is a representation of how many hours per week a student spends studying. Absences are a representation of how many days of school are missed in a school year, and will be used to try and find a potential correlation between days missed and GPA. Tutoring is a binary variable, and shows whether or not a student has participated in tutoring. This will be utilized to measure if GPA is most impacted by tutoring. Parental support describes how help from parents can impact a student's GPA. Found in the table of contents section of the dataset on Kaggle.com, this variable is measured on a scale of 0 to 4, with 0 being no support, and 4 being very high. Finally, GPA is the dependent variable that this project is trying to measure. It is a numerical showing of a student's academic performance, and is measured on a scale from 0 to 4, with 0 being the lowest and 4 being the highest.

Descriptive Statistics Table

For the variable Age, the descriptive statistics table (figure 1) shows a mean of 16.47 and a median of 16. This alludes to an even spread of students, which is represented in the bar chart for Age (figure 2). For StudyTime, the mean study hours is 9.77, with a high standard deviation of 5.65. This means that there is a high variation of study habits between the students. As seen in the histogram (figure 3), the bars representing study time per week are all pretty evenly distributed, besides a slight fall off as it reaches higher hours. When looking at Absences, there is a concerningly high mean of 14.54 absences per school year. Additionally, the high standard deviation of 8.47 is shown visually (figure 4) as the number of absences are spread out. However, this chart is very slightly skewed right, as there is a higher density of lower absences and a smaller amount of higher absences. Since tutoring is a binary variable, the 30.1% of students that have had a tutor is the mean of this variable. The median found in the descriptive statistics table is 0, which implies that more students did not receive tutoring. This is shown by the 69.9% in the pie chart (figure 5). The mean of parent support is 2.12, with a standard deviation of 1.12. This means that the graph is similar to a normal distribution. As seen in the histogram (figure 6), most students fall around the middle in terms of parental support. The GPA mean is 1.91, with a standard deviation of 0.915. Similar to the parental support chart, this

distribution follows a similar normal distribution curve. As shown in this histogram (figure 7), a bulk of the students are in the middle of the curve, with smaller amounts other the wider ends. Additionally the median of 1.89 compared to the mean of 1.91 suggests that the curve is somewhat symmetrical.

Model Specification

The general form of the model is as follows.

$$y = \beta_0 + \beta_1(x_1) + \beta_2(x_2) + \beta_3(x_3) + \beta_4(x_4) + \beta_5(x_5) + u, \text{ where:}$$

y = GPA

β_0 = intercept of the model

$\beta_1 - \beta_5$ = coefficients of the independent variables

$x_1 - x_5$ = independent variables

u = error

The aim of this model is to find out how GPA can be predicted using a variety of independent variables. The model intends to show how each variable impacts GPA, which can be further utilized to find out which variable factors into academic success the most. This will be done by doing regressions comparing GPA and each of the independent variables. **The final form of the model will be shown in the section “Final model”.** Moving on, the next section will discuss each independent variable and their causality.

Age: The first independent variable is age, and represents the students age in years. The causality of this variable is the belief that age may impact GPA due to maturity. Older students may have a greater understanding of how to balance their personal, social, and educational lives, while also having a greater motivation and academic purpose for college or trade school after high school.

Study Time: This variable is a representation of the amount of time (in hours) each student spends per week studying. The causality here is that more time spent studying will lead to a higher GPA, assuming study time is properly utilized for productive work.

Absences: This variable corresponds to the days of class missed. The causality of absences is that missing more days of class will lead to a lower GPA. This is because missing class can lead to the student having gaps in their knowledge of subjects, and lower participation and engagement, which can directly affect their test and homework scores, which will then impact GPA.

Tutoring: This variable describes whether or not students have received tutoring. It is a binary variable, where 1 = tutoring and 0 = no tutoring. The causality of tutoring is that tutoring will provide students with resources and extra practice to help master topics on exams and homeworks. In doing so, they will receive higher grades and have a higher GPA as a result.

Parental Support: This variable depicts how much support each student receives from their parents. According to the author of the dataset, the variable is set onto a four point scale with 0 being no support and 4 being the highest amount of support. The causality of this variable is that having a healthy learning environment at home will positively impact GPA.

Variable	Unit of measurement
Parental Support	Points
Tutoring	Yes/No
Study Time	Hours (per week)
Age	Years
Absences	Days (per school year)
GPA	Points

This next part will discuss variables that were considered but ultimately not used. As previously mentioned in the data section, the dataset pulled from kaggle.com included 16 variables. Three of the variables not included in this project were considered for the independent variables. These variables were gender, ethnicity, and extracurriculars. The variable “gender” would have been checked to see if gender correlates to higher GPA. While it is certainly possible for gender to impact GPA, the goal of this model is to focus on behaviors and decisions that impact GPA. More importantly however, gender is another binary variable. In order to keep the regressions as simple as possible, gender was removed from the data pool. Similar to gender, ethnicity likely has an impact on GPA. Different cultures may find education to be a higher priority, which in turn leads to the student getting more grades. However, due to the amount of dummy variables that would have needed to be created due to the different ethnicities, this variable was ruled out in order to keep the model as simple as possible. Extracurricular activities would have been seen if participation in extracurricular activities positively impacts GPA. However, this variable was ruled out for several reasons. First, it is likely that extracurricular activities have both positive and negative impacts on GPA. This is because while it may lead to better time management, it may also lead to time conflicts that take away time for studying. Additionally, adding another binary variable makes the model more complicated than it needs to be.

Correlations and Scatter Plots

The correlation matrix (figure 8) gives a breakdown of the relationships between the variables. Relationships between the independent variables and the dependent variables will be split into 3 categories (weak/no correlation, weak/moderate correlation, and high correlation).

Very Weak or No Correlation: Starting with the weakest correlation, age and GPA have a coefficient of 0.00028. This means that GPA is not impacted at all by age, showing that “maturity” among highschoolers likely does not impact academic performance.

Weak to Moderate Correlation: The lowest correlation in this category is tutoring and GPA with a positive correlation of 0.15. This means that students with tutoring do slightly better than students without tutoring. The next correlation is study time and GPA, which has a positive coefficient of 0.18. This alludes to how increases in study time have a small but greater impact on GPA than tutoring. The final relationship in this category is between parental support and GPA. With a positive correlation coefficient of 0.19, this means that parental support is a slightly more impactful variable to GPA than study time.

High Correlation: The final relationship is between absences and GPA. This relationship has a correlation coefficient of -0.92, which is extremely high. This means that there is an incredibly strong relationship between number of absences and GPA. More specifically, since it is a negative correlation, it implies that as more days of school are missed, the lower the students GPA will go.

The scatter plot for Age and GPA (figure 9) shows the data grouped into four vertical lines. Ultimately, no major patterns or trends exist due to the evenly distributed data, regardless of age. There is not much separation of data either as GPA has extreme variations. The only minor trend is that there is a fall off of students with extremely high and low GPA, which can be seen in the scatter plot as more data points become visible with less overlap on each data point. Each age group performs roughly the same, which aligns with the extremely low correlation coefficient of 0.00027 discussed in the correlation matrix (figure 8). Moving on to tutoring, from the pie chart (figure 5), it can be seen that less than one third of students use tutoring. The impact tutoring has had on overall GPA for the students that used it was positive, however, not extremely impactful compared to students who did not use tutoring. As shown in the scatter plot for tutoring and GPA (figure 10), there is a very small upward trend. Students who took tutoring not only have higher GPAs, but fewer students also have lower GPAs. The small trend also acknowledges the fairly weak correlation coefficient for this relationship, which is 0.15. Moving on, the study time and GPA scatterplot (figure 11) shows a very large spread of the data. However, there is a small trend as study time increases. This trend is that GPA increases as study time increases. Even though it is a large spread, the correlation coefficient of 0.18 shows that the aforementioned trend displays a slight correlation between GPA and study time per week. The parental support and GPA scatter plot (figure 12) follows a similar trend to the study time and GPA scatterplot. Here, the trend is a bit more noticeable, as the correlation coefficient is 0.19. GPA gradually increases as support increases. Similar to tutoring and study time, the positive correlation implies that this variable has a positive impact on GPA, however, it likely is not strong enough to impact GPA on its own. Finally, in the scatter plot between absences and GPA (figure 13), there is a clear correlation between the two variables. The decreasing trend as the number of absences increases means that this correlation is negative, which can also be proven by the correlation coefficient of -0.909. This coefficient is extremely strong, and is more than likely to be the most impactful variable on GPA out of the independent variables used in this project. The scatter plot shows an inverse relationship, where an increase in absences leads to a lower GPA.

Scatterplots between the independent variables (figures 14 - 23) all display extremely low correlation. They all lack any sort of noticeable trend or slope that would imply that there is correlation between them. This is further supported by the correlation matrix (figure 8), where it can be seen that the highest absolute value of correlations is 0.0358 between StudyTime and ParentalSupport.

Presentation of Regression Results/ Interpretation of the estimated coefficients

Starting off with Age and GPA, this regression (figure 24) has an R value (correlation coefficient) of 0.00028 and a P-value of 0.99 indicates there is virtually no linear relationship between the variables. The high P-value means the result is not statistically significant. The R^2 value of 0.0000000784 shows that the variance in GPA explained by Age is almost zero. Overall, Age has no explanatory or predictive value. The next regression is between GPA and Tutoring (figure 25). With an R value of 0.15 and P being $9.9e-13$, there is a weak but statistically significant positive linear relationship between the variables, as the P-value is far below 0.05. R^2 is 0.02250, which means about 2.25% of the variation in GPA is explained by Tutoring. Although the relationship is significant, the regression indicates a weak association. Moving on, the GPA and StudyTime regression (figure 26) displays $R = 0.18$ and $P = 2.2e-16$. This means there is a weak positive linear relationship that is also statistically significant due to the low P-value. The R^2 value is 0.0324, showing that about 3.24% of the variance in GPA is explained by StudyTime. This suggests that there is a small relationship between the variables. Next is the regression between GPA and Absences (figure 27). As shown in the figure, $R = -0.92$ and $P = 2.2e-16$, meaning there is a very strong negative linear relationship that is highly statistically significant. The R^2 value is 0.8464, which conveys that Absences explain 84.64% of the variance in GPA. This further supports the discussion of the scatterplot (figure 13) between these two variables that there is a strong relationship between them. The last regression is between GPA and Parental Support (figure 28). In this regression, $R = 0.19$ and $P = 2.2e-16$, displaying a weak positive linear relationship that is statistically significant. $R^2 = 0.03610$, indicating that 3.61% of the variation in GPA can be explained by Parental Support. Overall, there is a weak but significant relationship between the two variables.

Empirical Results

Variable regressed with GPA	P Value	Support/Reject Null Hypothesis	Statistically Significant?	Correlation Coefficient	Magnitude of coefficients	Functional form
Age	0.99	accept	No	0.00028	None	No relation
Tutoring	$9.9e-15$	accept	Yes	0.15	weak	positive
Study Time	$2.2e-16$	reject	Yes	0.18	weak	positive
Absences	$2.2e-16$	reject	Yes	-0.92	strong	negative
Parental Support	$2.2e-16$	reject	yes	0.19	weak	positive

Final model

$$y = 2.857 - 0.01123(x_1) + 0.2596(x_2) + 0.0287(x_3) - 0.0994(x_4) + 0.152(x_5)$$

(0.0745) (0.0044) (0.0106) (0.00088) (0.00058) (0.0044)

y = GPA

x₁ = Age, x₂ = Tutoring, x₃ = Study Time, x₄ = Absences, x₅ = Parent Support

Parenthesis below model = standard error of each variable

*coefficients and robust std. error based on coefficient test (figure 46)

Discussion of the Assumptions of Multiple Linear Regression Analysis**1. Linearity**

For the GPA and Age linearity check (figure 29), there is no sloped line, indications there is no linear relationship between GPA and Age. Next is GPA and Tutoring (figure 30), which has a very subtle slope, meaning that there is a very small linear relationship between GPA and Tutoring. With GPA and StudyTime (figure 31), there is no sloped line. Similar to GPA and Age, this means there is no linear relationship between the two variables. The GPA and Absences linearity check (figure 32) shows a very steep line, implying a high negative linear relationship. Lastly, GPA and Parental Support shows a modest linear relationship (figure 33).

2. Independence

The fitted x residual graph of the linear regression model shows that there is a randomized spread of values (figure 34). This is further supported by the Durbin-Watson Test statistic of 1.96 (figure 35). This value is well within the necessary bounds of 1.5 - 2.5. Additionally, since it is relatively close to 2.0, this entails that the model has little to no autocorrelation and the error terms are independent.

3. Multicollinearity

Multicollinearity has already been disproven through the correlation matrix (figure 8). However, it can be further emphasized through the Variance Inflation Factor (VIF) Test (figure 35). Notably, all values are no greater than 1.0022, meaning that multicollinearity is not present.

4. Heteroskedasticity

As seen in the fitted x residual graph of the model (figure 34), there is no significant spread in error terms. Most of the terms fall between 0.5 and -0.5 on the y axis, and this remains constant as x moves across the graph. This is further supported by the Breusch-Pagan test (figure 36) as the p value is 0.91. This means the null hypothesis that there is no heteroscedasticity is accepted.

5. Normality

The histogram with normal curve (figure 37) and density plot (figure 38) of the model both show normality within the model. In the histogram, most of the data falls under the normal curve, with the excess in the center. The density plot furthers this notion as it also resembles a normal distribution curve. Normality can be further proven by the QQ plot (figure 39) as the plot shows the data almost lining up perfectly with the 45 degree line. The Shapiro-Wik test was also

undertaken to prove normality (figure 40). The test shows a p value of 0.27, significantly greater than 0.05, as well as a W statistic of 0.999, meaning the distribution is not significantly different from a normal distribution.

6. Endogeneity

Endogeneity was tested by plotting error(residual) terms alongside independent variables. These plots (figures 41 - 45) show no significant trends or curves. This ultimately means that the independent variables are not correlated to the error terms, disproving any endogeneity in the model.

Conclusion

To summarize the results, correlation coefficients of each variable with GPA are as follows. Age is 0.00028, tutoring is 0.15, study time is 0.18, absences is -0.92, and parent support is 0.19. Additionally, the model passed all assumptions of linear regressions. This implies that the model can be used for prediction and reasoning within reason. From the results, it can also be implied and predicted that Absences have the greatest impact on GPA. Tutoring, study time, and parental support also impact GPA but at a slightly lesser level. Finally, age does not seem to have any impact on GPA. Future research could expand outwards into many other educational sectors. For example, college GPA could be predicted using variables such as study time and absences. However, other variables could be included such as class difficulty or major. Research could also expand in the opposite direction of development. For example, this model could serve as a baseline for early childhood education, where new variables such as time spent doing extracurricular activities, or days spent in learning centers can be utilized to predict childhood development.

Note from Author

This dataset about highschool students' GPA is the second dataset selected for this project. The original dataset was a dataset involving NBA players and their salary, position, and personal stats such as points, assists, and rebounds, with the goal of the model seeing what position's performance on the court best represents their salary. The reason why the project was transitioned away from this dataset was because of the position variable. Unlike sports like soccer where players can be further simplified to offensive and defensive positions, basketball can only be simplified to 5. These positions are point guard, shooting guard, small forward, power forward, and center. This means that 5 new dummy variables would have to be created. Thus, a new dataset was found with no dummy variables to simplify the project.

Appendix A: Charts / Graphs

Figure 1 - Descriptive Statistics Table

```
> summary(Age)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 15.00  15.00  16.00  16.47  17.00  18.00

> summary(StudyTime)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.001057  5.043079  9.705363  9.771992 14.408410 19.978094

> summary(Absences)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 0.00   7.00  15.00  14.54  22.00  29.00

> summary(HasTutoring)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000  0.0000  0.0000  0.3014  1.0000  1.0000

> summary(ParentSupportLevel)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 0.000  1.000  2.000  2.122  3.000  4.000

> summary(GPA)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 0.000  1.175  1.893  1.906  2.622  4.000

> sd(Age)
[1] 1.123798
> sd(StudyTime)
[1] 5.652774
> sd(Absences)
[1] 8.467417
> sd(HasTutoring)
[1] 0.4589712
> sd(ParentSupportLevel)
[1] 1.122813
> sd(GPA)
[1] 0.9151558
```

Figure 2 - Age Barchart

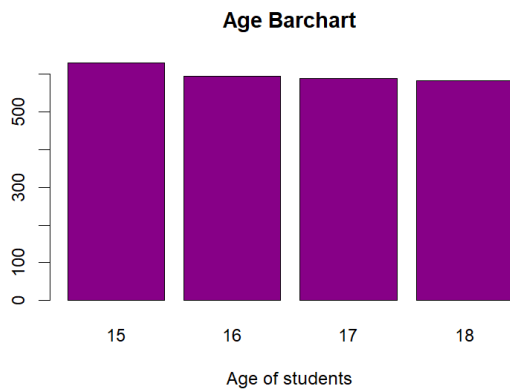


Figure 3 - Study Time Histogram

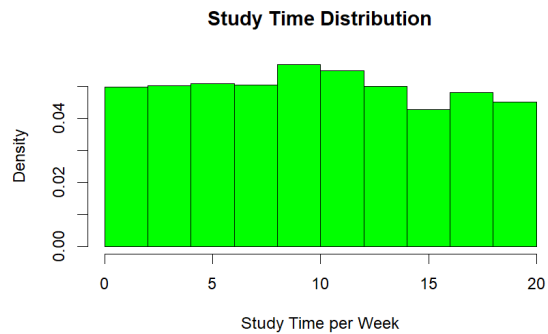


Figure 4 - Absences Histogram

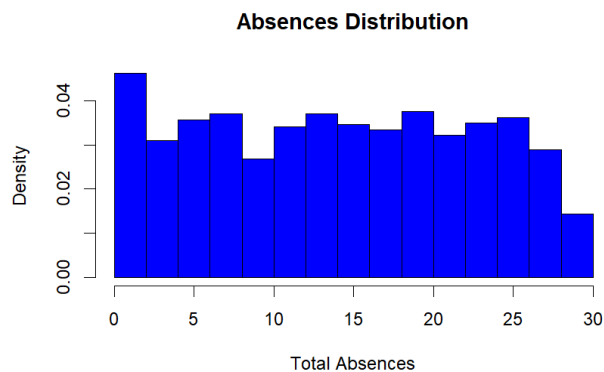


Figure 5 - Tutoring Pie Chart

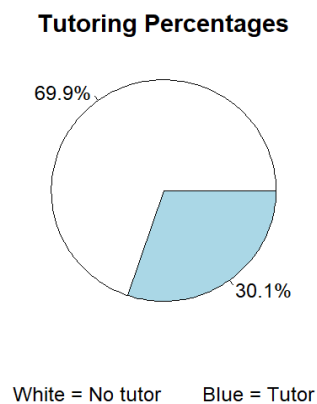


Figure 6 - Parent Support Histogram

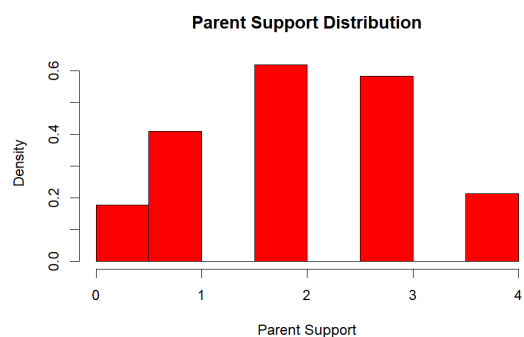


Figure 7 - GPA Histogram

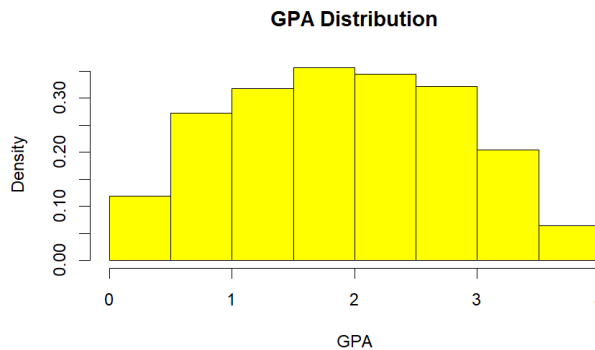


Figure 8 - Correlation Matrix

	Age	StudyTimeWeekly	Absences	Tutoring
Age	1.0000000000	-0.006800031	-0.011510913	-0.0120764764
StudyTimeWeekly	-0.0068000307	1.0000000000	0.009325535	0.0289302963
Absences	-0.0115109127	0.009325535	1.0000000000	-0.0155336258
Tutoring	-0.0120764764	0.028930296	-0.015533626	1.0000000000
ParentalSupport	0.0331970528	0.035799964	0.002107808	-0.0008237873
GPA	0.0002753882	0.179275127	-0.919313576	0.1451189200

	ParentalSupport	GPA
Age	0.0331970528	0.0002753882
StudyTimeWeekly	0.0357999635	0.1792751269
Absences	0.0021078077	-0.9193135764
Tutoring	-0.0008237873	0.1451189200
ParentalSupport	1.0000000000	0.1907737278
GPA	0.1907737278	1.0000000000

Figure 9 - Age/GPA Scatterplot

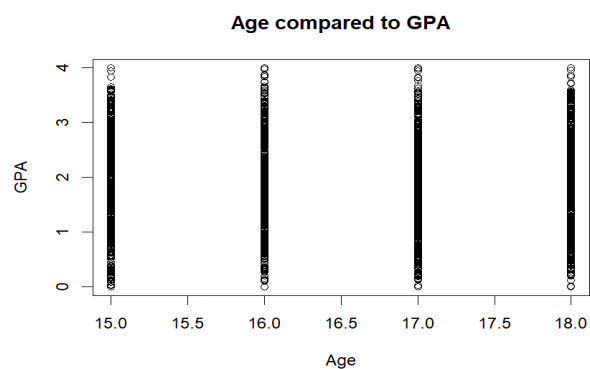


Figure 10 - Tutoring/GPA Scatterplot

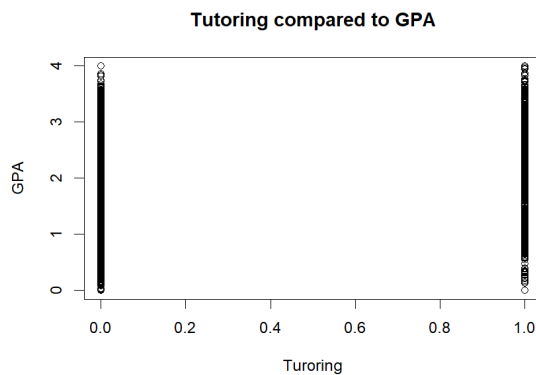


Figure 11 - Study Time/GPA Scatterplot

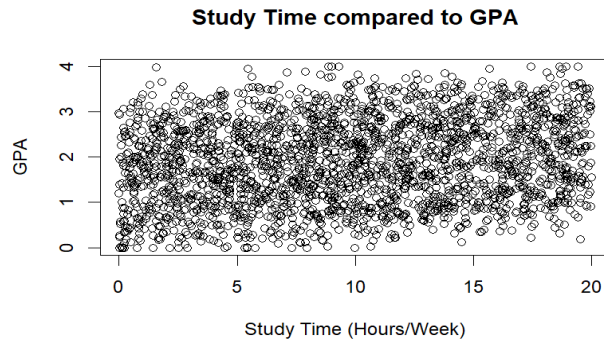


Figure 12 - Parent Support/GPA Scatterplot

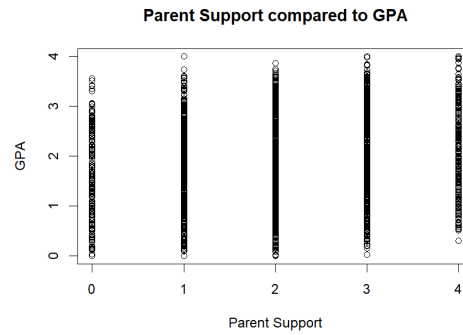


Figure 13 - Absences/GPA Scatterplot

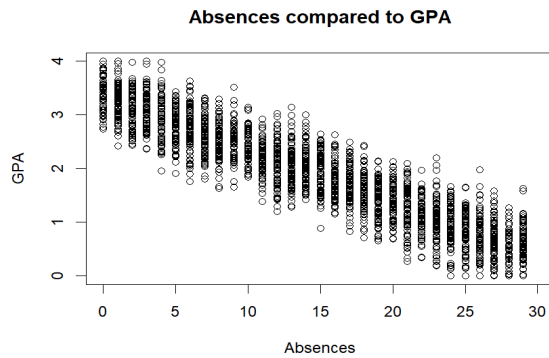


Figure 14 - Age/Absences Scatterplot

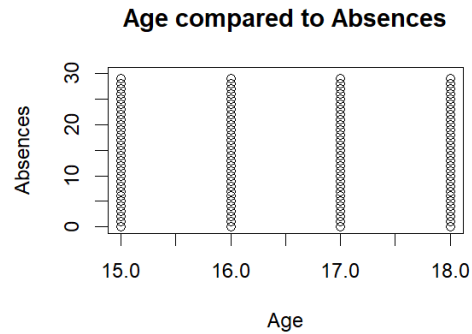


Figure 15 - Age/Study Time Scatterplot

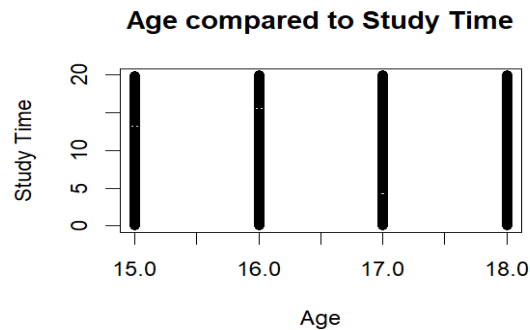


Figure 16 - Age/Tutoring Scatterplot

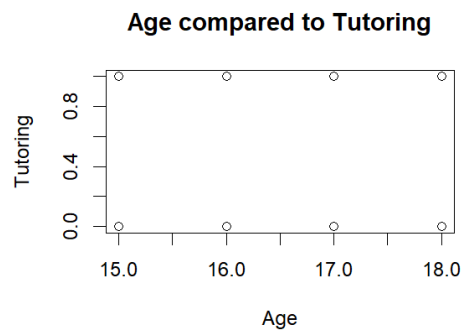


Figure 17 - Age/Parental Support Scatterplot

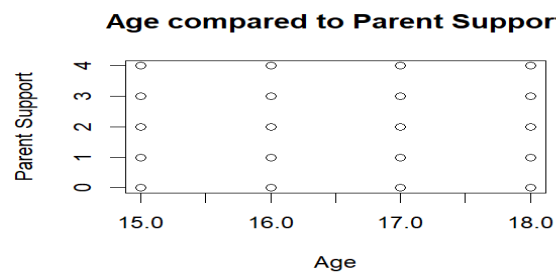


Figure 18 - StudyTime/Absences Scatterplot

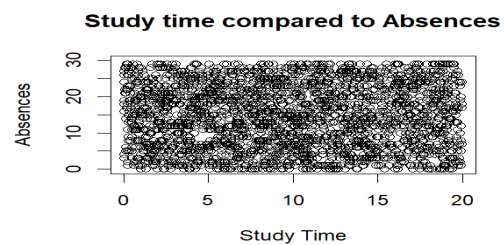


Figure 19 - Study Time/Tutoring Scatterplot

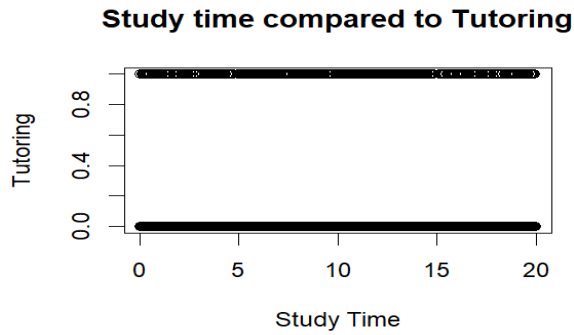


Figure 20 - StudyTime/Parent Support Scatter

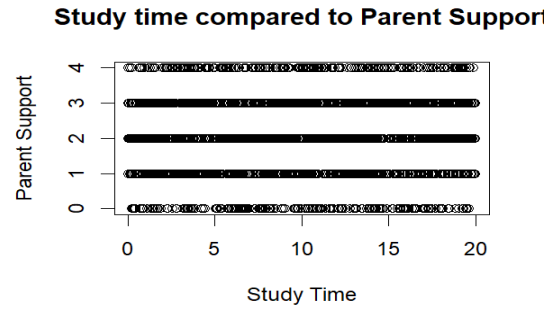


Figure 21 - Absences/Tutoring Scatterplot

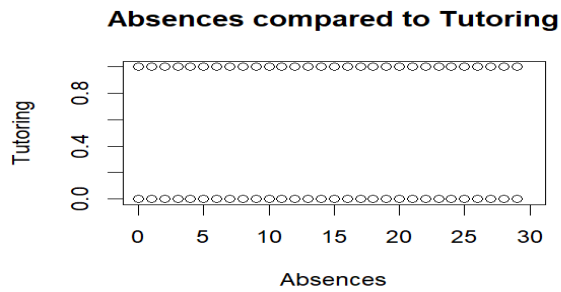


Figure 22 - Absences/Parental Support Scatterplot

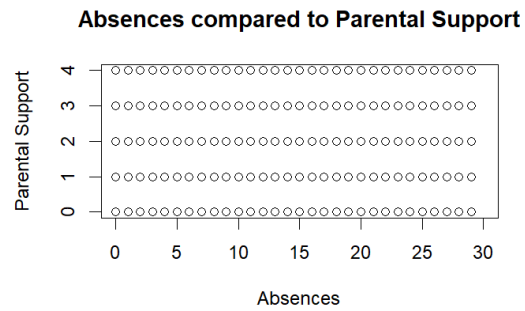


Figure 23 - Tutoring/Parental Support Scatterplot

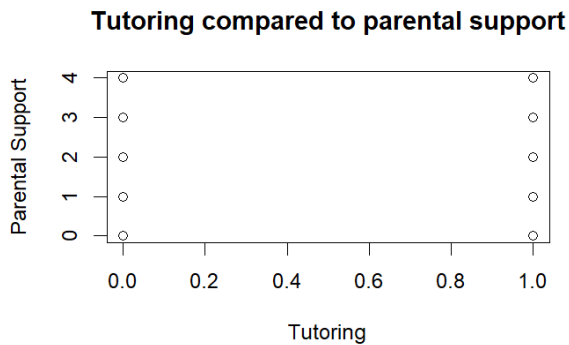


Figure 24 - GPA/Age Regression

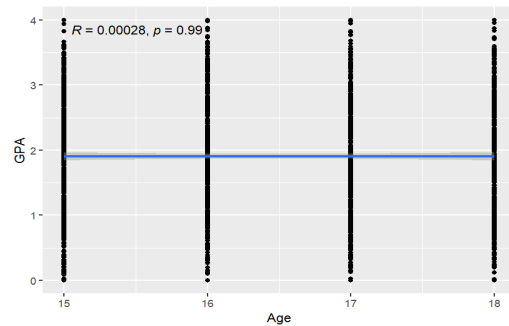


Figure 25 - GPA/Tutoring Regression

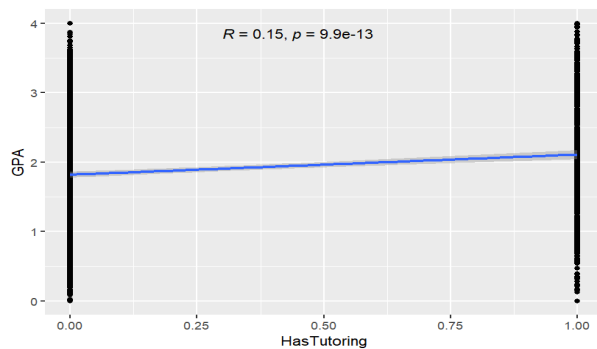


Figure 26 - GPA/StudyTime Regression

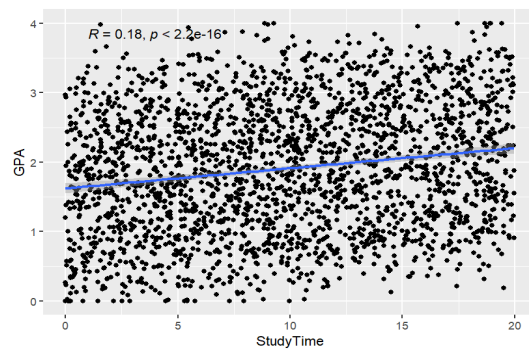


Figure 27 - GPA/Absences Regression

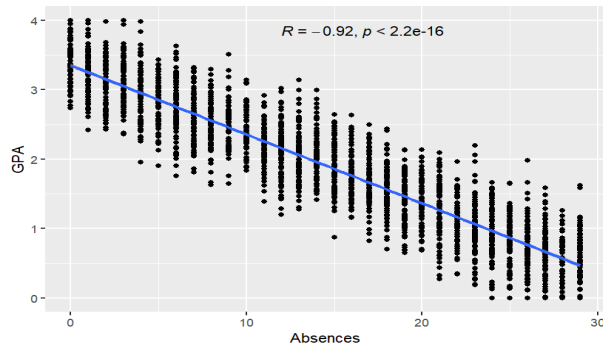


Figure 28 - GPA/Parent Support Regression

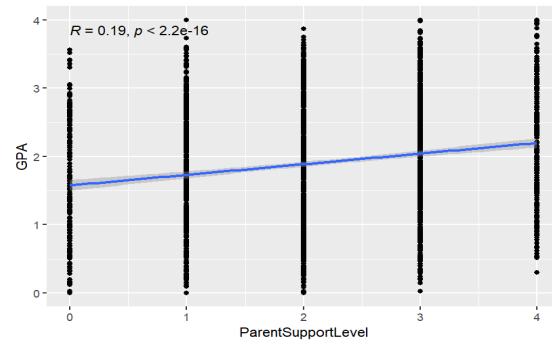


Figure 29 - GPA/Age Linearity

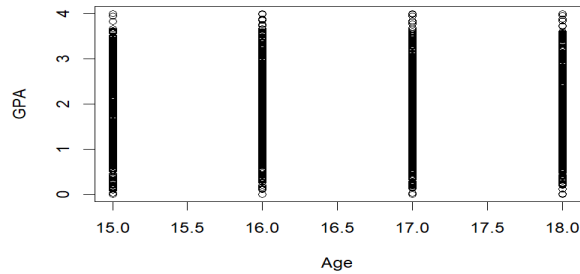


Figure 30 - GPA/Tutoring Linearity

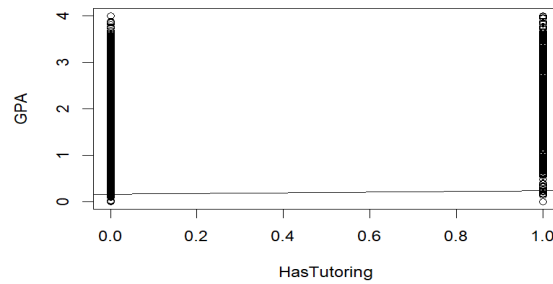


Figure 31 - GPA/StudyTime Linearity

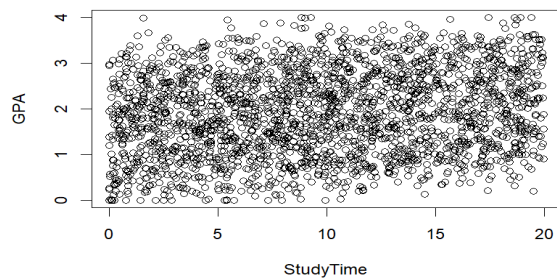


Figure 32 - GPA/Absences Linearity

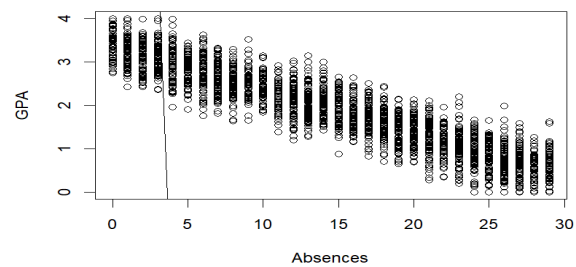


Figure 33 - GPA/Parent Support Linearity

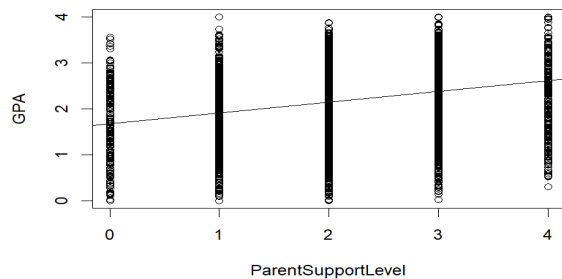


Figure 34 - Fitted x Residual of Model

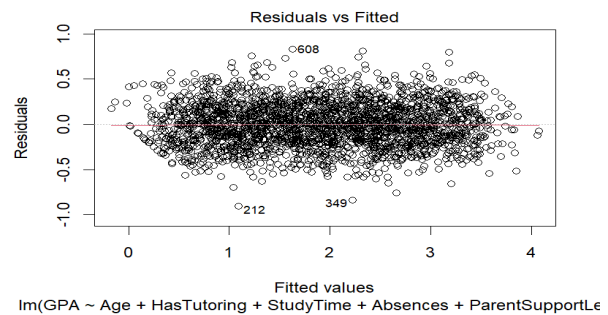


Figure 35 - DurbinWatson and VIF

```

> durbinWatsonTest(model)
lag Autocorrelation D-W Statistic p-value
1 0.01937068 1.961031 0.312
Alternative hypothesis: rho != 0
> vif(model)
Age HasTutoring StudyTime Absences
1.001445 1.001236 1.002276 1.000477
ParentSupportLevel
1.002412

```

Figure 36 - BP Test

studentized Breusch-Pagan test

```

data: model
BP = 1.4637, df = 5, p-value = 0.9172

```

Figure 37 - Histogram with normal curve

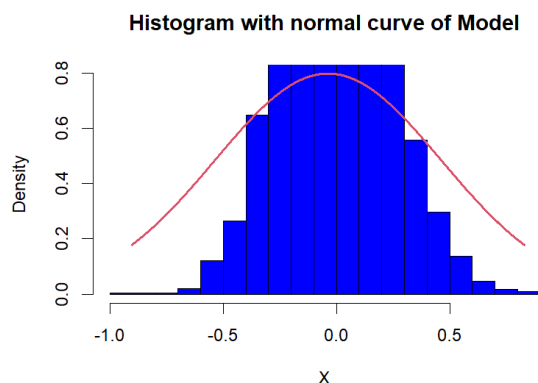


Figure 38 - Density Plot

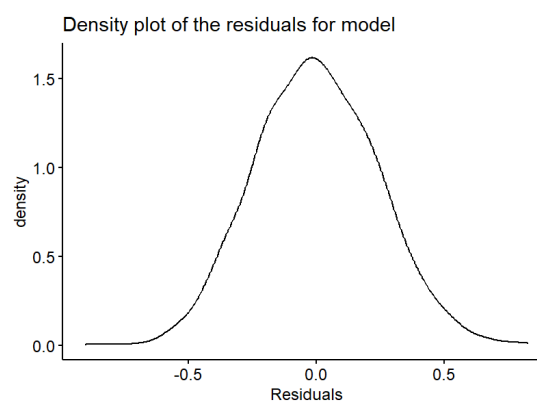


Figure 39 - QQ Plot

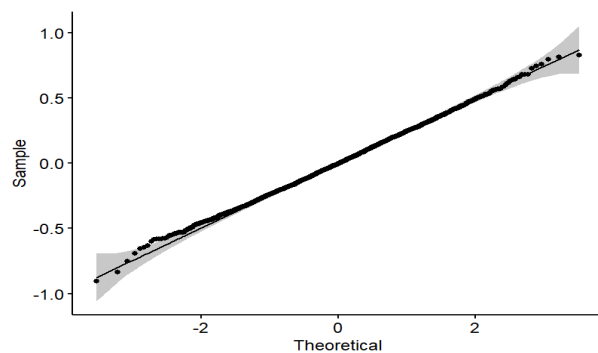


Figure 40 - Shapiro Normality Test

Shapiro-Wilk normality test

```

data: den1
W = 0.99909, p-value = 0.2701

```

Figure 41 - Residuals vs Age

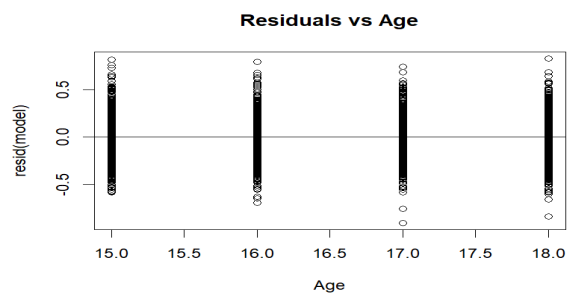


Figure 42 - Residuals vs Tutoring

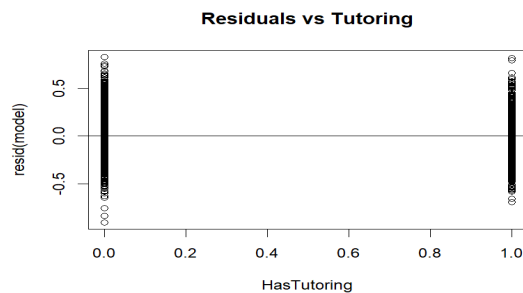


Figure 43 - Residuals vs StudyTime

Figure 44 - Residuals vs Absences

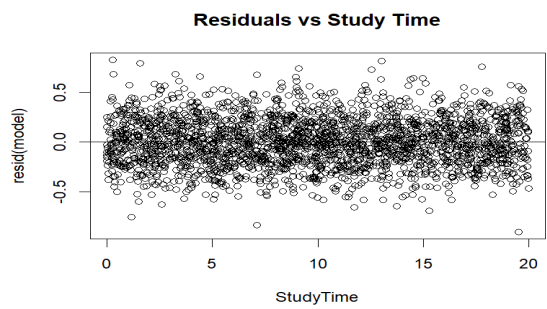


Figure 45 - Residuals vs Parental Support

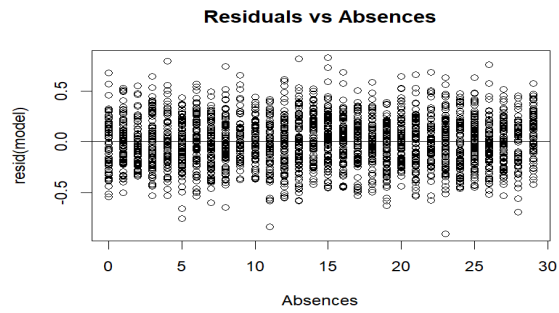
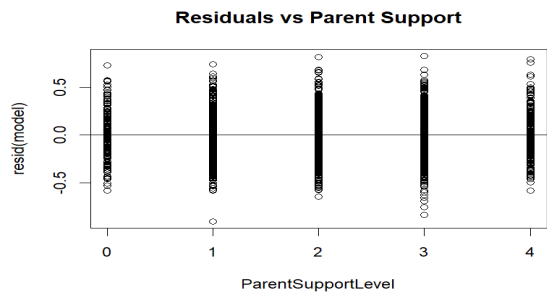


Figure 46 - Coefficients and Std. Errors



t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	2.85681201	0.07454264	38.3245	< 2e-16	***
Age	-0.01123020	0.00443756	-2.5307	0.01145	*
HasTutoring	0.25061499	0.01059999	23.6429	< 2e-16	***
StudyTime	0.02872484	0.00087825	32.7068	< 2e-16	***
Absences	-0.09938668	0.00057652	-172.3919	< 2e-16	***
ParentSupportLevel	0.15235149	0.00442629	34.4196	< 2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Appendix B - RStudio Code

```
require(ggplot2)
install.packages("tidyverse")
install.packages("readxl")
library(readxl)
library(geomtextpath)
library(tidyverse)
library(ggpubr)
library(corrplot)

library("ggplot2") # Data visualization
library("dplyr") # Data manipulation
install.packages("kableExtra")
install.packages("moments")
install.packages("fUnitRoots") #For analyzing trends and unit roots in financial time series, the
augmented Dickey-Fuller Test
install.packages('car')
install.packages("lmtest")
install.packages("ecm") # For the Durbin H test
library(fUnitRoots) #For analyzing trends and unit roots
library("readxl") #The readxl package makes it easy to get data out of Excel and into R
library(moments)
library(car)
library(lmtest)
library(dplyr)
library(haven)
library(lmtest) # To calculate robust standard errors, we can use the coeftest() function from
the lmtest package
library(sandwich) # and the vcovHC() function from the sandwich package
library(mFilter) # Filtering the data
library(ecm) # For the Durbin H test

StudentData <- read_xlsx("C:/Users/garre/Downloads/BUS110/StudentData.xlsx")

Age <- StudentData$Age
StudyTime <- StudentData$StudyTimeWeekly
Absences <- StudentData$Absences
HasTutoring <- StudentData$Tutoring
ParentSupportLevel <- StudentData$ParentalSupport
GPA <- StudentData$GPA
YesTut <- subset(StudentData, HasTutoring == 1)
NoTut <- subset(StudentData, HasTutoring != 1)
```



```
table(Age)
```

```
barplot(table(Age),  
  main = "Age Barchart",  
  xlab= "Age of students",  
  col= "darkmagenta")
```

```
table(StudyTime)
```

```
hist(StudyTime,  
  main = "Study Time Distribution",  
  xlab= "Study Time per Week",  
  col= "green",  
  freq=FALSE)
```

```
hist(Absences,  
  main = "Absences Distribution",  
  xlab= "Total Absences",  
  col= "blue",  
  freq=FALSE)
```

```
TutTab = table(HasTutoring)
```

```
TutorProb = TutTab/nrow(StudentData)*100
```

```
TutPie = pie(TutTab,  
  labels = paste(round(TutorProb, 1), "%", sep = ""),  
  radius = 1,  
  main = "Tutoring Percentages",  
  xlab = "White = No tutor    Blue = Tutor")
```

```
hist(ParentSupportLevel,  
  main = "Parent Support Distribution",  
  xlab = "Parent Support",  
  col = "red",  
  freq = FALSE)
```

```
hist(GPA,  
  main = "GPA Distribution",  
  xlab = "GPA",  
  col = "yellow",  
  freq = FALSE)
```

```
plot(x = Age, y = GPA,
     xlab = "Age",
     ylab = "GPA",
     xlim = c(15, 18),
     ylim = c(0, 4),
     main = "Age compared to GPA"
)
```

```
plot(x = StudyTime, y = GPA,
     xlab = "Study Time (Hours/Week)",
     ylab = "GPA",
     xlim = c(0, 20),
     ylim = c(0, 4),
     main = "Study Time compared to GPA"
)
```

```
plot(x = Absences, y = GPA,
     xlab = "Absences",
     ylab = "GPA",
     xlim = c(0, 30),
     ylim = c(0, 4),
     main = "Absences compared to GPA"
)
```

```
plot(x = ParentSupportLevel, y = GPA,
     xlab = "Parent Support",
     ylab = "GPA",
     xlim = c(0, 4),
     ylim = c(0, 4),
     main = "Parent Support compared to GPA"
)
```

```
TutTab = table(HasTutoring)
TutorProb = TutTab/nrow(StudentData)*100
```

```
TutPie = pie(TutTab,
             labels = paste(round(TutorProb, 1), "%", sep = ""),
             radius = 1,
             main = "Tutoring Percentages",
             xlab = "White = No tutor    Blue = Tutor")
```

```
plot(x = HasTutoring, y = GPA,
     xlab = "Tutoring",
     ylab = "GPA",
```

```
xlim = c(0, 1),
ylim = c(0, 4),
main = "Tutoring compared to GPA"
)

cor_matrix <- cor(StudentData)

print(cor_matrix)

plot(x = Age, y = Absences,
     xlab = "Age",
     ylab = "Absences",
     xlim = c(15, 18),
     ylim = c(0, 30),
     main = "Age compared to Absences"
)

plot(x = Age, y = StudyTime,
     xlab = "Age",
     ylab = "Study Time",
     xlim = c(15, 18),
     ylim = c(0, 20),
     main = "Age compared to Study Time"
)

plot(x = Age, y = HasTutoring,
     xlab = "Age",
     ylab = "Tutoring",
     xlim = c(15, 18),
     ylim = c(0, 1),
     main = "Age compared to Tutoring"
)

plot(x = Age, y = ParentSupportLevel,
     xlab = "Age",
     ylab = "Parent Support",
     xlim = c(15, 18),
     ylim = c(0, 4),
     main = "Age compared to Parent Support"
)

plot(x = StudyTime, y = Absences,
     xlab = "Study Time",
     ylab = "Absences",
```

```
xlim = c(0, 20),
ylim = c(0, 30),
main = "Study time compared to Absences"
)

plot(x = StudyTime, y = HasTutoring,
     xlab = "Study Time",
     ylab = "Tutoring",
     xlim = c(0, 20),
     ylim = c(0, 1),
     main = "Study time compared to Tutoring"
)

plot(x = StudyTime, y = ParentSupportLevel,
     xlab = "Study Time",
     ylab = "Parent Support",
     xlim = c(0, 20),
     ylim = c(0, 4),
     main = "Study time compared to Parent Support"
)

plot(x = Absences, y = HasTutoring,
     xlab = "Absences",
     ylab = "Tutoring",
     xlim = c(0, 30),
     ylim = c(0, 1),
     main = "Absences compared to Tutoring"
)

plot(x = Absences, y = ParentSupportLevel,
     xlab = "Absences",
     ylab = "Parental Support",
     xlim = c(0, 30),
     ylim = c(0, 4),
     main = "Absences compared to Parental Support"
)

plot(x = HasTutoring, y = ParentSupportLevel,
     xlab = "Tutoring",
     ylab = "Parental Support",
     xlim = c(0, 1),
     ylim = c(0, 4),
     main = "Tutoring compared to parental support"
)
```

```
#regressions
```

```
A <- ggplot(StudentData, aes(Age, GPA)) +  
  geom_point() +  
  geom_smooth(method = lm) +  
  stat_cor(method = "pearson", label.x = 15.05)
```

A

```
HT <- ggplot(StudentData, aes(HasTutoring, GPA)) +  
  geom_point() +  
  geom_smooth(method = lm) +  
  stat_cor(method = "pearson", label.x = 0.3)
```

HT

```
ST <- ggplot(StudentData, aes(StudyTime, GPA)) +  
  geom_point() +  
  geom_smooth(method = lm) +  
  stat_cor(method = "pearson", label.x = 1)
```

ST

```
AB <- ggplot(StudentData, aes(Absences, GPA)) +  
  geom_point() +  
  geom_smooth(method = lm) +  
  stat_cor(method = "pearson", label.x = 12)
```

AB

```
P <- ggplot(StudentData, aes(ParentSupportLevel, GPA)) +  
  geom_point() +  
  geom_smooth(method = lm) +  
  stat_cor(method = "pearson", label.x = 0)
```

P

```
#assumptions
```

```
#linearity
```

```
plot(Age, GPA) +  
  abline(lm(Age ~ GPA))
```

```
plot(HasTutoring, GPA) +  
  abline(lm(HasTutoring ~ GPA))
```

```
plot(StudyTime, GPA) +  
  abline(lm(StudyTime ~ GPA))
```

```

plot(Absences, GPA) +
  abline(lm(Absences ~ GPA))

plot(ParentSupportLevel, GPA) +
  abline(lm(ParentSupportLevel ~ GPA))

#independence / multicollinearity
model <- lm(GPA ~ Age + HasTutoring + StudyTime + Absences + ParentSupportLevel)
summary(model)
durbinWatsonTest(model)
# Variance Inflation Factor
vif(model)

plot (fitted.values(model), residuals(model))
#add a horizontal line at y=0
abline(0, 0)

#heteroskedasticity
plot(model, which = 1, main = "")
bp_test1 <- bptest(model)
print(bp_test1)

#Normality
X <- resid(model)
hist(X, prob = TRUE)
# X-axis grid
X2 <- seq(min(X), max(X), length = 2392)
# Normal curve
fun <- dnorm(X2, mean = mean(X2), sd = sd(X2))
# Histogram
hist(X, prob = TRUE, col = "blue",
     ylim = c(0, max(fun)),
     main = "Histogram with normal curve of Model")
lines(X2, fun, col = 2, lwd = 2)

#Density Plot
den1 <- residuals(model)
# Density plot: the density plot provides a visual judgment about whether the distribution is bell
shaped.
hist(den1)
ggdensity(den1,
  main = "Density plot of the residuals for model",
  xlab = "Residuals")

```

```
#QQ Plot/ Shapiro Test
ggqqplot(den1)
shapiro.test(den1)
hist(g, density=20, breaks=20, prob=TRUE,
     xlab="x-variable", ylim=c(0, 2),
     main="normal curve over histogram")
curve(dnorm(x = 1, mean=m, sd=std),
     col="darkblue", lwd=2, add=TRUE, yaxt="n")

#endogeneity
plot(Age, resid(model), main = "Residuals vs Age")
abline(0,0)
plot(HasTutoring, resid(model), main = "Residuals vs Tutoring")
abline(0,0)
plot(StudyTime, resid(model), main = "Residuals vs Study Time")
abline(0,0)
plot(Absences, resid(model), main = "Residuals vs Absences")
abline(0,0)
plot(ParentSupportLevel, resid(model), main = "Residuals vs Parent Support")
abline(0,0)

#robust standard errors
coeftest(model, vcov = vcovHC(model, type = 'HC0'))
```